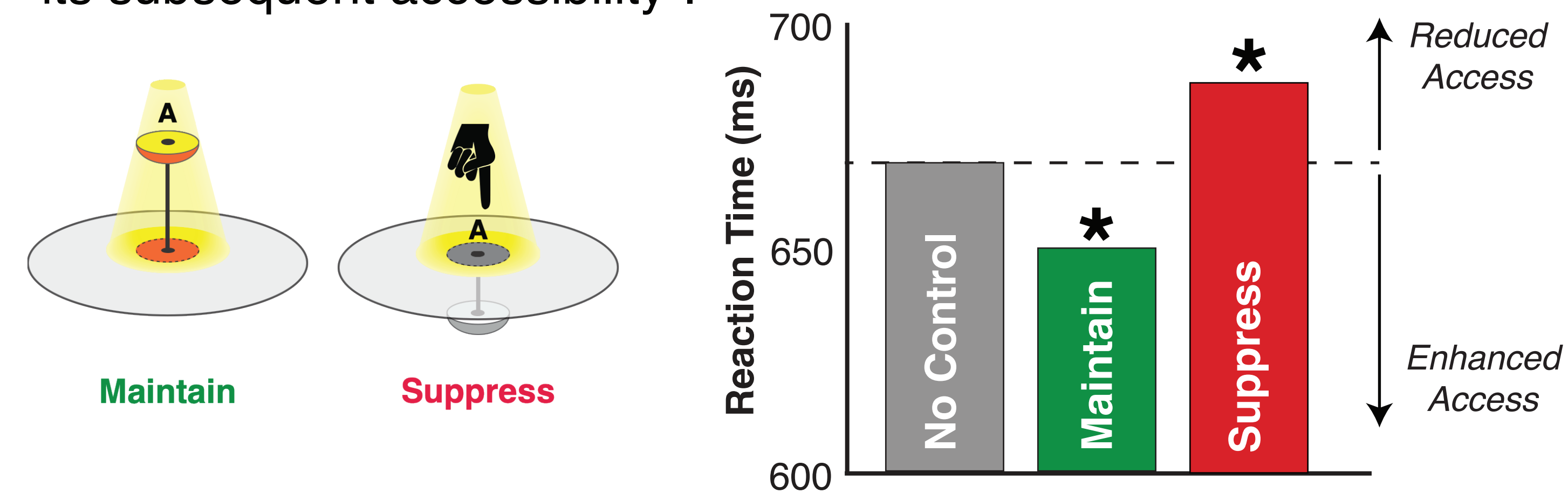


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<sup>a</sup>University of Texas at Austin; <sup>b</sup>University of Colorado Boulder

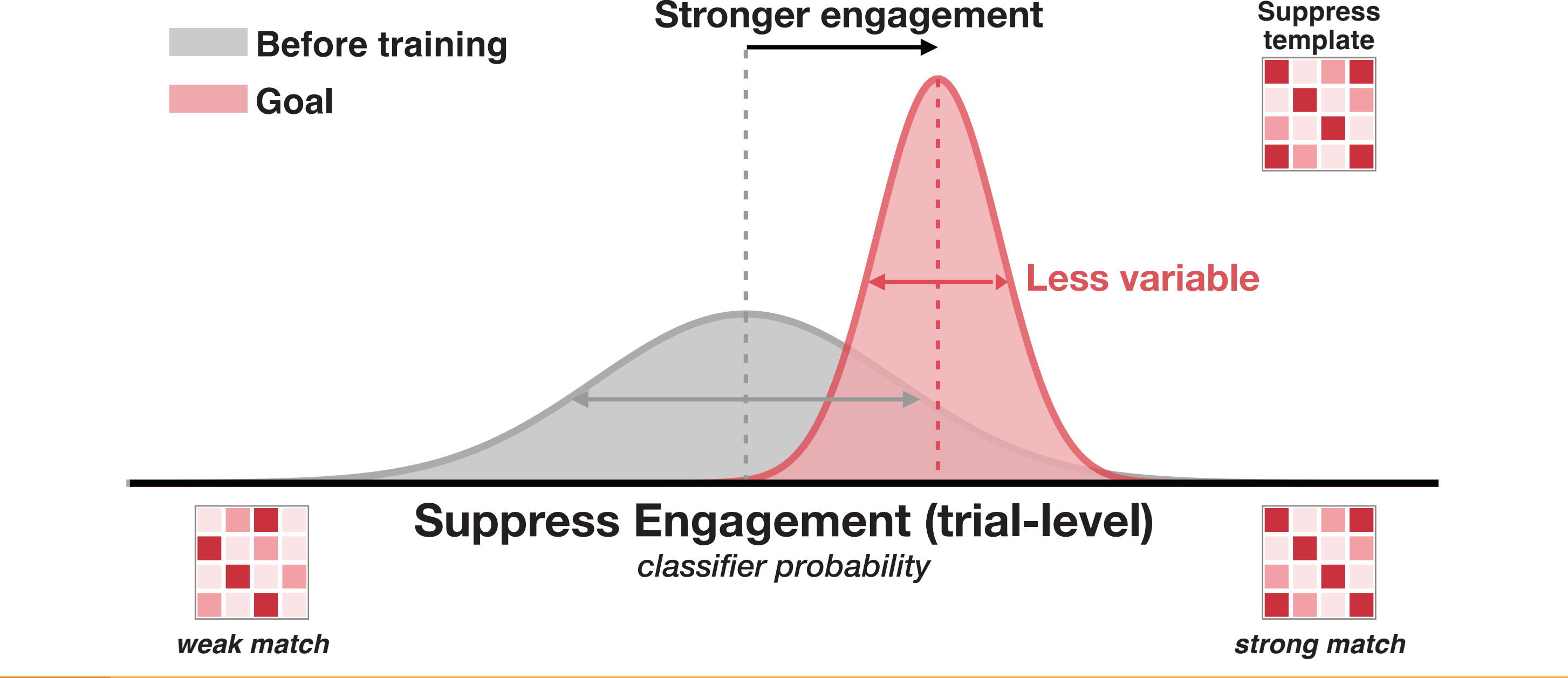
## 1 INTRODUCTION

### Background

- Directly suppressing unwanted working memory content reduces its subsequent accessibility<sup>1</sup>.
- Suppression engagement fluctuates trial to trial<sup>2</sup>, and weaker engagement predicts difficulty controlling unwanted thoughts<sup>3</sup>.
- Because suppression is a complex, internal process, individuals lack a direct signal of how well they are engaging it.
- Decoded neurofeedback can provide this signal by tracking the distributed activity patterns of suppression engagement<sup>4</sup>.



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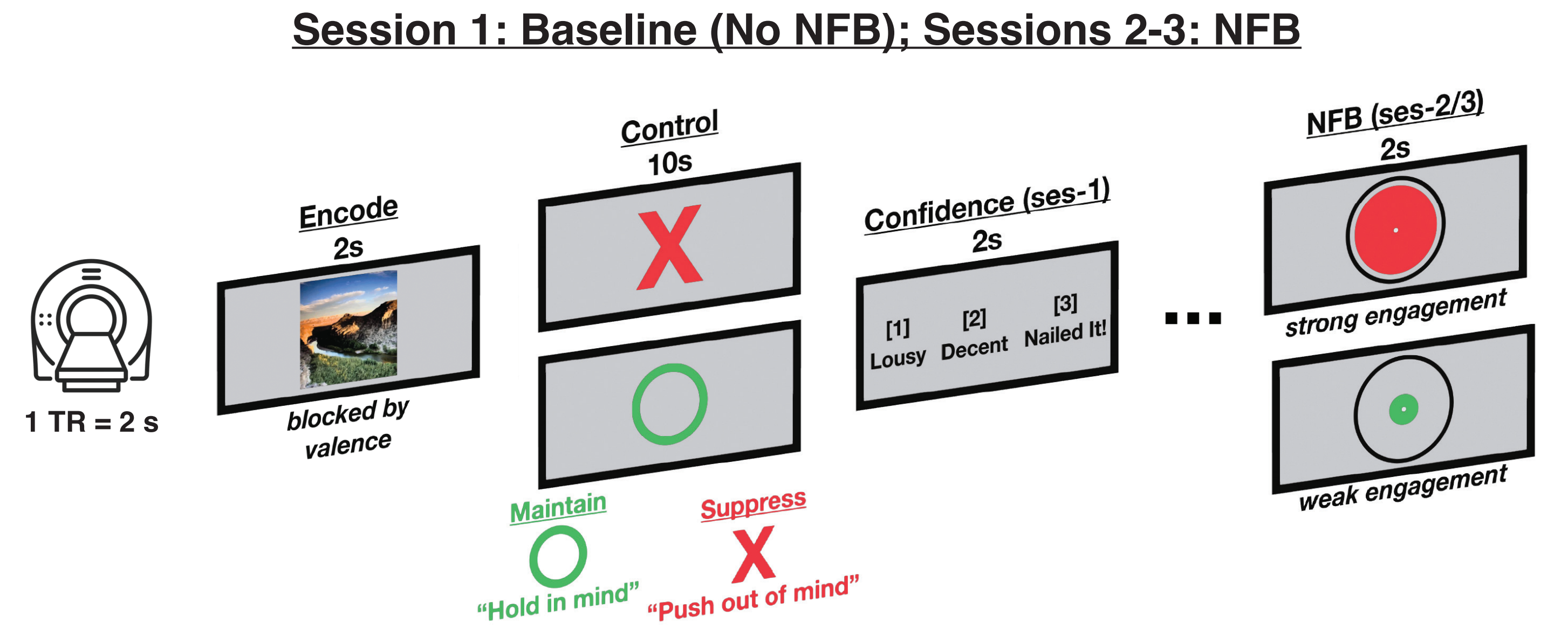


### RESEARCH QUESTION

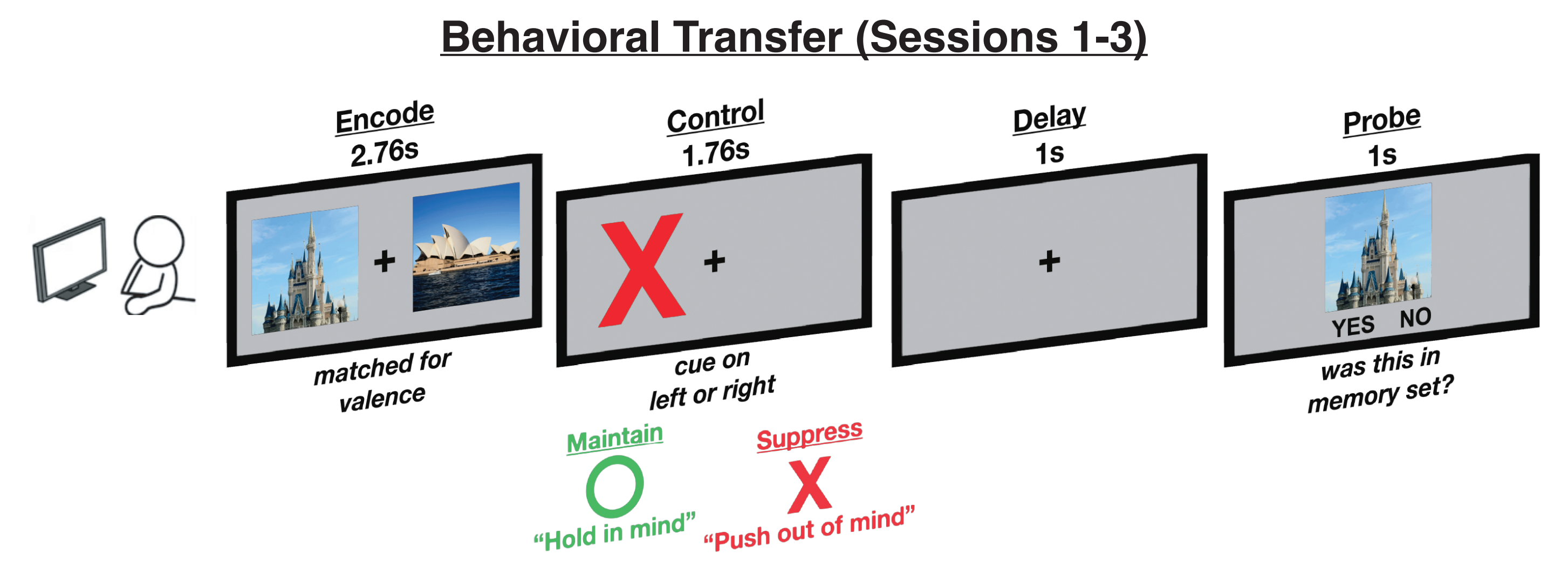
Can decoded neurofeedback increase the strength and consistency of suppression engagement?

## 2 EXPERIMENTAL APPROACH (N = 16)

H1: Neurofeedback will strengthen suppression engagement and make it more consistent across training sessions.

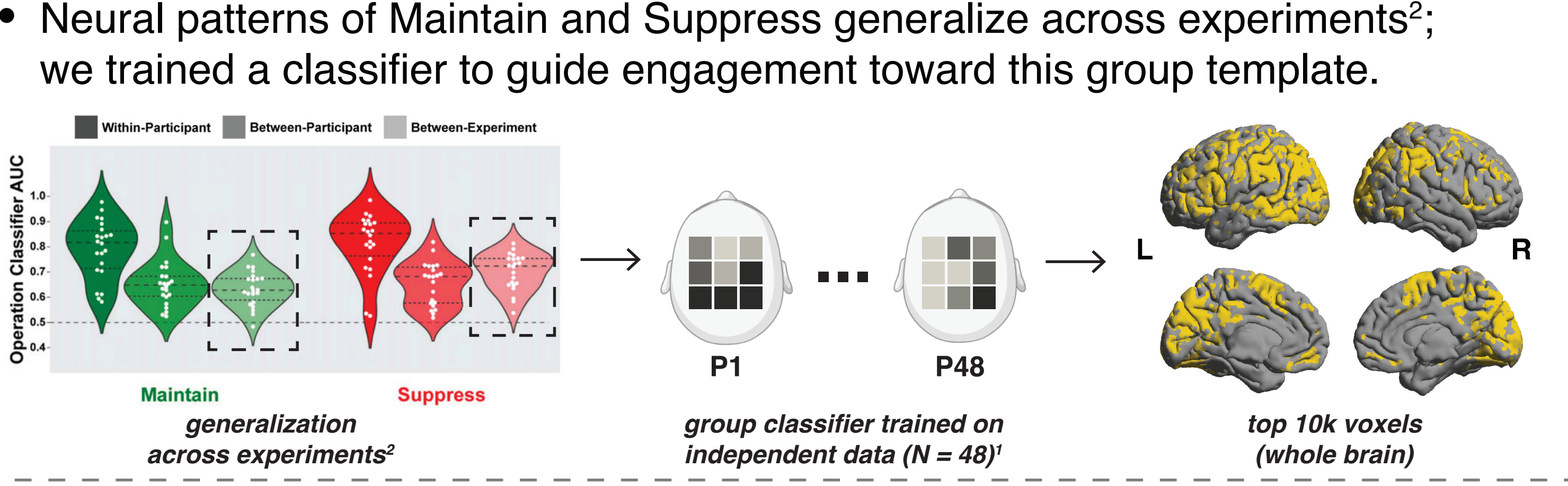


H2: These neural changes will transfer to behavior, reducing the accessibility of suppressed memories.

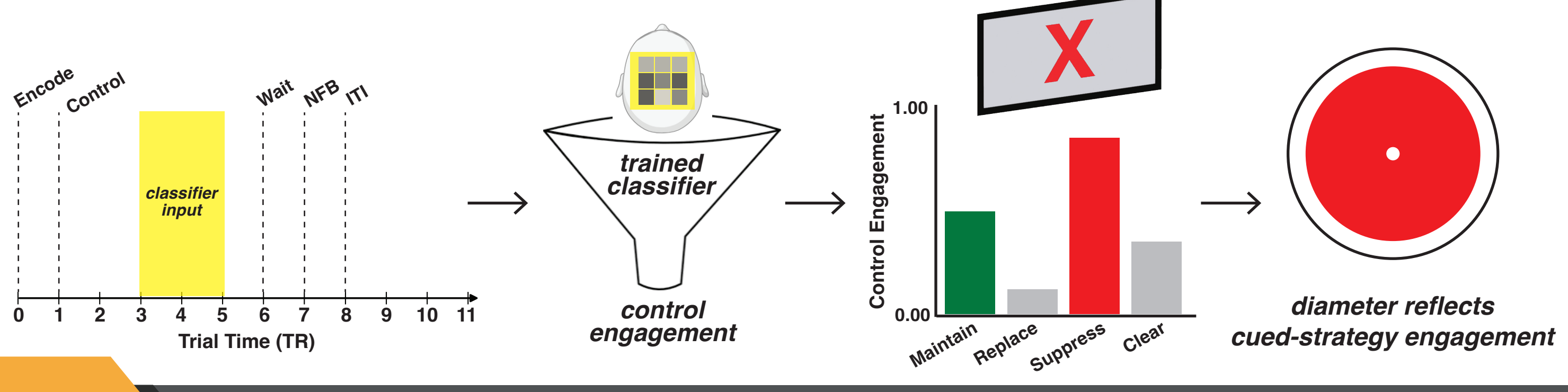


## 3 DECODED NEUROFEEDBACK

### Classifier Training

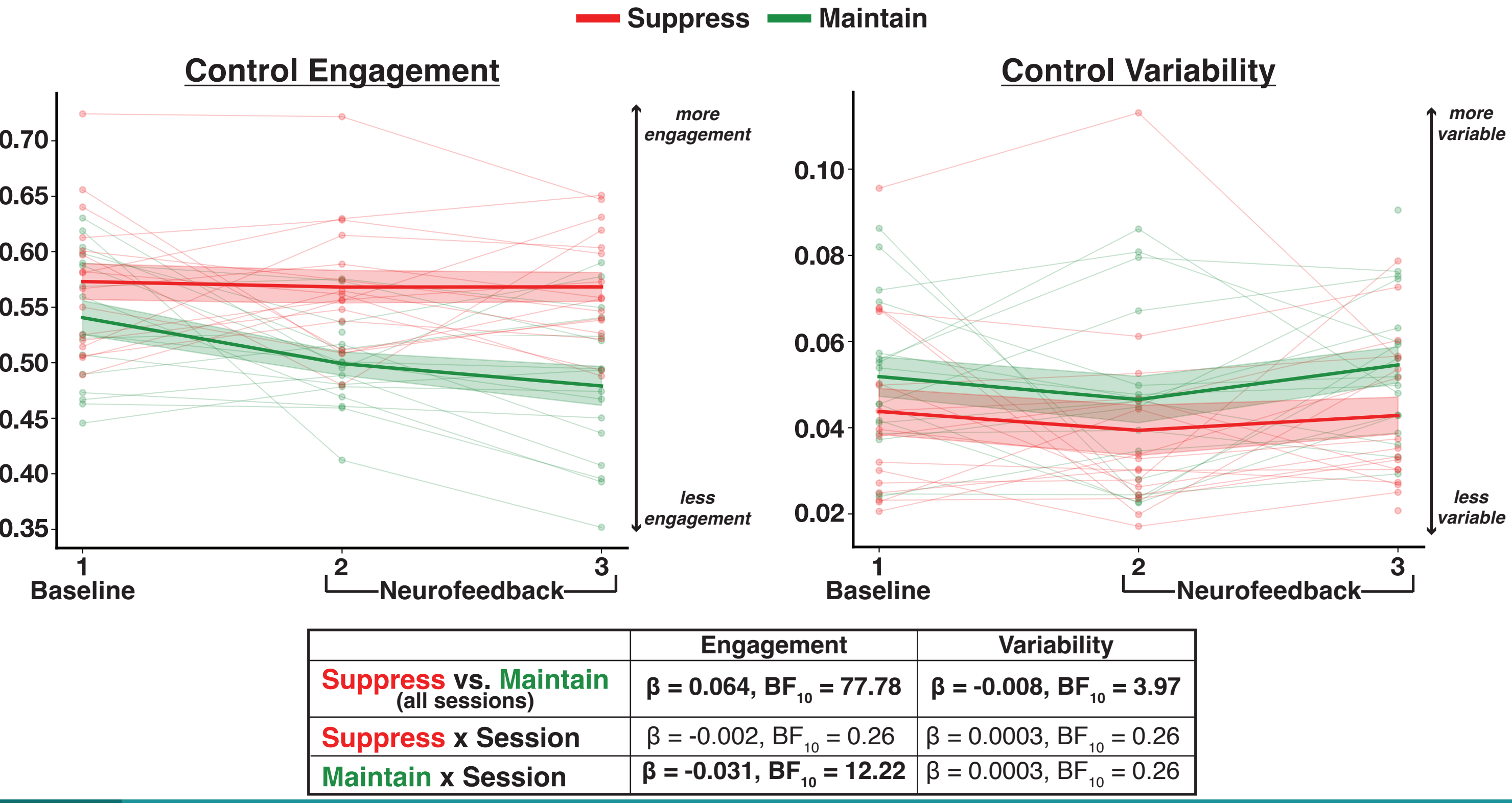


### Closed-Loop Decoding

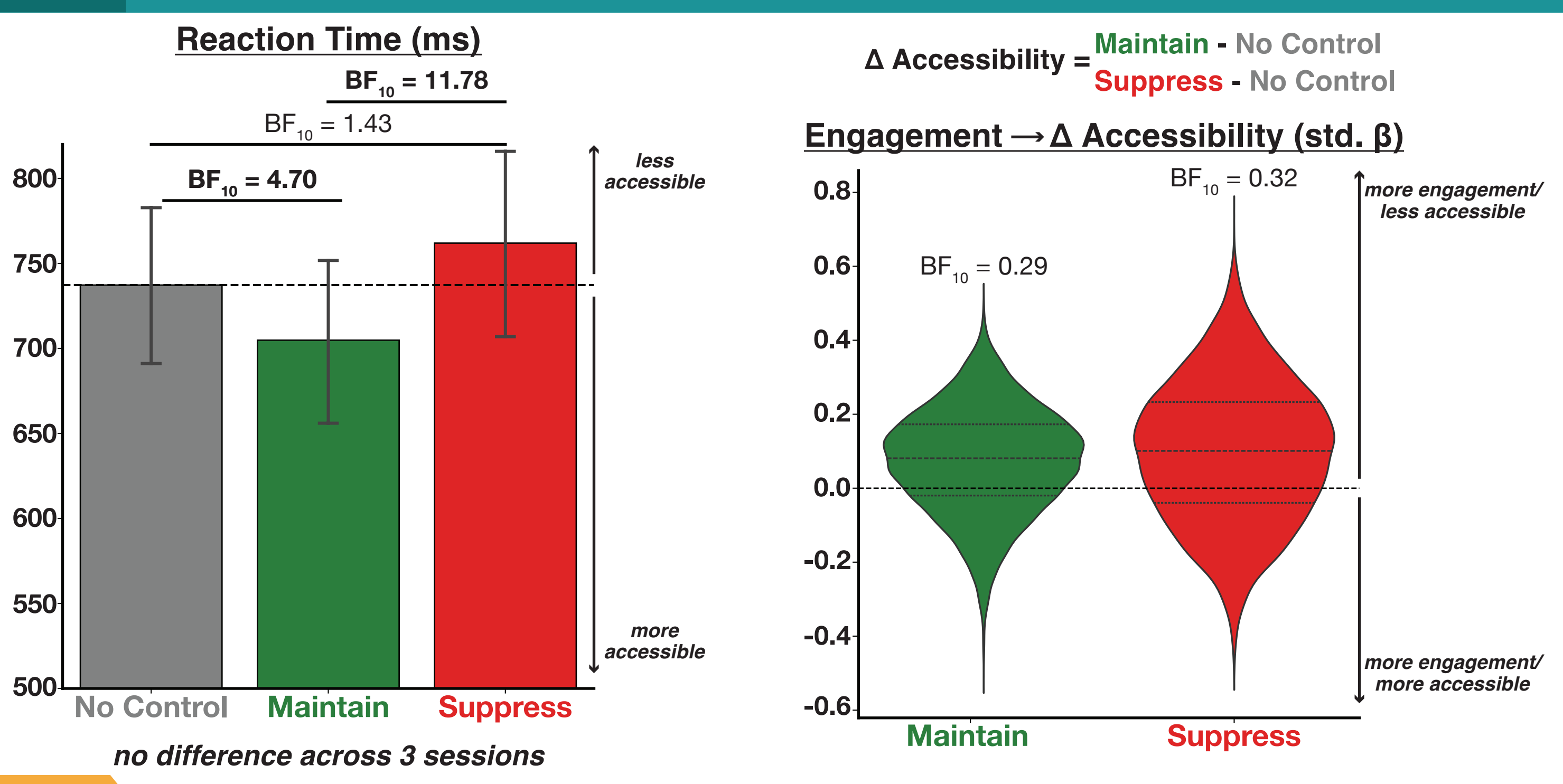


## 4 RESULTS

### Suppression engagement is stable across training



### Control engagement does not predict behavioral accessibility



## 5 DISCUSSION

- Neurofeedback did not modulate suppression engagement or consistency, and did not predict changes in memory accessibility.
- Engagement may index, but not drive, downstream changes: manipulating even the strongest signal need not change behavior<sup>5,6</sup>.
- Engagement is only one piece: how it shapes the targeted memory's representation may better index whether control succeeds.

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